PROJECT SHOWCASE / ENGINEERING & PRODUCT PORTFOLIO

Complete Project Portfolio

Product & Business Briefs

Six projects: MetricForge · Petinha · Gig Goal Tracker · Bills Reminder · MathQuest Arcade · Career Town (CV RPG)

Prepared for professional review | November 4, 2026 | English edition

Evidence note: Based on prior project discussions, described architecture, selected development/debugging history and portfolio records available at preparation time. This is a portfolio case study, not a fresh source-code audit, external security review, market study or verified commercial due diligence. Planned capabilities and unverified outcomes are labeled accordingly.

Contents

- 1. MetricForge - Social analytics SaaS
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How to use this dossier

For hiring discussions, use the technical edition to examine component responsibilities, design decisions, implementation evidence and areas worth demonstrating in code. For product or sponsorship meetings, use the business edition to evaluate problem framing, proposed value, distribution hypotheses, validation and principal risks. The corresponding per-project PDF can be attached to each project page.

MetricForge | Product & Business Brief

Social analytics SaaS | Product stage: Working application with YouTube integration; broader platform scope evolving

Product premise

MetricForge is a multi-user social analytics platform intended to centralize account data, media performance, and analytical insights across social channels. Its confirmed development history includes authentication, a protected analytics dashboard, GraphQL domain models, MongoDB persistence, and YouTube account authorization and synchronization.

Customer problem

Creators and small organizations must move between platform dashboards to correlate audience, content, engagement, and performance. Context becomes fragmented, comparison is expensive, and historical snapshots are difficult to standardize.

Target users

Independent creators, agencies, social-media operators and small multi-channel teams.

Value proposition

One workspace for channel observations and actionable reporting instead of repeated manual review.

Potential differentiation

Potential differentiation through opinionated dashboards, content-level analysis, accessible reporting and workflow simplicity. Not a proven unique advantage.

Alternatives and competitive pressure

Native YouTube Studio and other network dashboards, plus established social analytics suites; product validation must demonstrate why a user would switch or add another paid tool.

Commercial model hypothesis

Candidate freemium or trial with paid per-workspace subscription, gated by number of connected accounts, synchronization frequency and deeper analytics. Prices are hypotheses, not existing plans with billing proof.

Initial go-to-market experiment

Recruit early YouTube creators; offer guided onboarding and request comparative feedback on reporting time saved.

Validation plan and decision metrics

- 10-15 structured interviews; identify one must-have weekly use case.
- Recruit 5-10 active pilot users who connect their own channels.

- Measure weekly active use, completed syncs, data freshness, report exports and willingness to pay.
- Do not pitch revenue or multi-platform coverage without auditable proof.

Commercial and operational risks

- Provider API policy/quota changes and OAuth verification.
- Privacy and security exposure when managing social access tokens.
- Competing with established analytics products; customer acquisition cost.
- Ambiguous attribution and platform-metric comparability.

Phased go-to-market roadmap

- Pilot with one well-supported provider and a narrow dashboard.
- Validate retention and operational costs of synchronization.
- Expand export/reporting and permissions before additional social integrations.
- Introduce billing only after real customer evidence.

Sponsor / investor diligence checklist

- Demonstrable product build and concise feature tour.
- Evidence of actual use: users, retention and repeated workflows rather than feature count alone.
- Costs, pricing assumptions and support effort; no unsupported revenue forecast.
- IP ownership, licensed assets, privacy commitments and provider API dependencies.
- Specific requested partnership: pilot access, introductions, development support or investment, plus milestones.

Important: All market positioning and monetization options in this brief are hypotheses to validate. No TAM, revenue, valuation, user growth, partnerships or investment return is represented as verified.

Petinha | Product & Business Brief

Virtual pet / interactive web game | Product stage: Existing application under active engine migration and movement/animation debugging

Product premise

Petinha is an interactive virtual-pet experience with state-based needs and player actions. Development discussions cover a Next.js application backed by MongoDB and authentication, while the game runtime is being migrated toward Phaser 3 so movement, animation and object interactions are owned by a coherent game loop.

Customer problem

Interactive virtual pets must consistently coordinate persistent needs, character animations, inventory-like actions and scene interactions. Mixing DOM/CSS animations with game-engine transforms can introduce divergent state and unpredictable movement.

Target users

Casual game players, families and users seeking a lightweight browser pet experience.

Value proposition

A persistent, playful pet relationship with short interactive sessions and collectible/progression potential.

Potential differentiation

Character identity, approachable visual design and browser access could create a distinct experience; retention is unproven.

Alternatives and competitive pressure

Established virtual-pet apps, casual mobile games and free browser games.

Commercial model hypothesis

Possible cosmetic purchases or a premium content pack, with free core play; not implemented or validated.

Initial go-to-market experiment

Invite a small test audience, observe first-session completion and return visits, then evaluate organic sharing.

Validation plan and decision metrics

- Test a playable 10-minute core loop with at least 10 users.
- Measure tutorial completion, day-1/day-7 retention and meaningful sessions per user.
- Ask whether players understand pet feedback and action consequences.

- Avoid monetization until the technical core loop is reliable.

Commercial and operational risks

- High content demand and difficulty retaining casual players.
- Animation and game-state bugs disrupting progression.
- Potential child-directed audience considerations, privacy and content safeguards.
- Costs of new artwork and ongoing gameplay content.

Phased go-to-market roadmap

- Stabilize core pet loop and Phaser migration.
- Release a controlled public demo.
- Instrument retention and usability.
- Add content or cosmetic monetization only after behavioral evidence.

Sponsor / investor diligence checklist

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Gig Goal Tracker | Product & Business Brief

Gig-work earnings productivity | Product stage: Actively developed earnings/goal tracking app; advanced mileage/tax reporting scoped or evolving

Product premise

Gig Goal Tracker is a mobile-friendly web application to help gig workers manage earning targets, trips and work sessions. It is grounded in the developer's direct use of gig delivery workflows and aims to keep essential daily tracking simple while progressively revealing advanced analysis.

Customer problem

Gig workers often evaluate offers, maintain earnings goals and assemble tax records across fragmented tools. Without a consistent session and trip history, it is difficult to interpret work performance or reconcile mileage.

Target users

Delivery drivers and other independent gig workers who need easy earnings tracking.

Value proposition

Clear earning targets, fewer manual calculations, and a usable history of work and mileage.

Potential differentiation

Firsthand gig-work workflow, simplicity by default, and optional advanced reports rather than a crowded interface.

Alternatives and competitive pressure

Spreadsheets, platform earnings pages and dedicated mileage/expense tracking apps.

Commercial model hypothesis

Previously considered beta invitation with 7/14/30 days of free access, then a simple membership (illustrative $5.99/month or $49.99/year). These are tentative, not verified sales.

Initial go-to-market experiment

Pilot with local driver communities, word of mouth, invited users and worker-focused demonstrations.

Validation plan and decision metrics

- Invite 15-30 drivers into a private beta.
- Measure 7-day activation, weekly data entry completion, repeat use and churn.
- Test willingness to pay against free spreadsheets and existing tracker tools.
- Confirm subscription economics with support costs and willingness-to-pay interviews.

Commercial and operational risks

- High price sensitivity and abandonment from manual entry friction.
- Privacy obligations around earnings and location/mileage information.
- Confusion between records and tax advice.
- Changes to gig platform workflows and app-store/web discovery.

Phased go-to-market roadmap

- Polish daily goal and trip logging.
- Add optional odometer + reconciliation and export.
- Publish reliable statistics and beta onboarding.
- Add billing after measured retention and engagement.

Sponsor / investor diligence checklist

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Bills Reminder | Product & Business Brief

Recurring payment organization | Product stage: Backend/data model and reminder scheduling worked on; complete feature parity unverified

Product premise

Bills Reminder is a personal payment-organization application intended to consolidate recurring bill schedules, payment status and proactive notifications. Development history includes MongoDB Atlas collections, payment records, notification logs and work on job locking for scheduled processing.

Customer problem

Recurring bills vary in frequency, due date and status. Manual calendars and disconnected inbox reminders can lead to missed payments, poor visibility and duplicate alerts.

Target users

Individuals, households and freelancers with recurring personal obligations.

Value proposition

One payment calendar with timely reminders and a comprehensible paid/unpaid history.

Potential differentiation

Potentially lightweight, transparent recurring-bill experience with configurable summaries; advantages require user testing.

Alternatives and competitive pressure

Bank bill alerts, calendar reminders, budgeting apps and personal finance suites.

Commercial model hypothesis

Freemium with optional premium reminders, household collaboration or advanced reporting; hypothesis only.

Initial go-to-market experiment

Test with users managing diverse bill types; focus on email digest usefulness rather than broad finance branding.

Validation plan and decision metrics

- Interview households and freelancers about missed/late-payment workflows.
- Pilot reminders with clear consent and opt-out.
- Measure reminder usefulness, false alarms, and manually confirmed bill payments.
- Estimate delivery costs and subscriber willingness to pay.

Commercial and operational risks

- Highly crowded consumer utility category and low switching costs.
- Trust and notification accuracy when dates or status change.
- Email delivery reputation and anti-spam compliance.
- Avoid collection of banking credentials unless strictly required and secure.

Phased go-to-market roadmap

- Prove recurring bill accuracy and scheduler safety.
- Ship simple email reminder and payment confirmation flow.
- Collect retention and alert effectiveness data.
- Expand to household/reporting features only when requested.

Sponsor / investor diligence checklist

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MathQuest Arcade | Product & Business Brief

Educational logic / math games | Product stage: Playable mini-game work and extensive visual design; full catalog and polish evolving

Product premise

MathQuest Arcade is a web portal for mathematical reasoning and logic mini-games, built around an illustrated medieval tavern hub. Users select an avatar and access experiences such as Cows and Bulls and Hidden Tiles. The project emphasizes approachable learning-by-doing and a coherent visual world.

Customer problem

Static math practice can feel abstract and repetitive. A game-based collection may encourage exploration, iteration and reasoning through bite-sized challenges without requiring a long learning session.

Target users

Families, students, casual puzzle players and potentially educators.

Value proposition

Quick entertaining practice of logic, numerical deduction and mathematical concepts.

Potential differentiation

A consistent RPG-inspired world connecting multiple short logic games, bilingual access and friendly art direction.

Alternatives and competitive pressure

Free educational game sites, browser puzzles, mobile brain-training products and classroom platforms.

Commercial model hypothesis

Free basic play with optional premium game packs, educational licenses or merchandise; none should be presented as validated revenue.

Initial go-to-market experiment

Share a browser demo with families and teachers, test whether a second game is discovered after the first.

Validation plan and decision metrics

- Observe players of different ages complete onboarding without help.
- Measure game-start to completion, repeat visits and crossover to other games.
- Request educator feedback on how each rule maps to a learning objective.
- Run small controlled tests before claiming improvements in mathematical proficiency.

Commercial and operational risks

- High demand for new levels/assets and competition with free games.
- Protection of children’s data and parental controls when relevant.
- Accessibility and clarity for younger players.
- Potential intellectual property review for third-party assets or copied mechanics/art.

Phased go-to-market roadmap

- Release 1-2 polished mini-games and mobile-first tutorials.
- Add telemetry, accessibility improvements and language QA.
- Expand games after repeat-use evidence.
- Explore educator offerings if educational outcomes are substantiated.

Sponsor / investor diligence checklist

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Career Town (CV RPG) | Product & Business Brief

Interactive resume / portfolio | Product stage: Concept, world assets, Tiled configuration and character designs; complete public game unverified

Product premise

Career Town is an interactive RPG-style professional portfolio in which a recruiter explores a small town. The candidate’s education, skills, work and projects appear as discoverable buildings instead of being confined to a static resume page.

Customer problem

Traditional portfolios place many technologies and project details into long scrollable pages. A visually navigable professional map can create stronger differentiation and invite exploration, but must retain straightforward accessibility for busy reviewers.

Target users

Primarily employers, recruiters and collaborators; secondarily possible clients interested in custom interactive portfolios.

Value proposition

A distinctive demonstration of front-end, game interaction, UX and visual storytelling skills.

Potential differentiation

An authored RPG town linking real professional evidence to an explorable interface; its core value is credibility and memorability.

Alternatives and competitive pressure

Conventional personal sites, template portfolios, interactive CVs and online game-like portfolios.

Commercial model hypothesis

Primarily a career-marketing asset. A possible future service/productization of interactive portfolios is speculative and not the current business model.

Initial go-to-market experiment

Send directly with job applications and professional outreach; compare performance against a conventional portfolio.

Validation plan and decision metrics

- Recruiters must find important information quickly without instructions.
- Track project-detail visits and resume downloads without excessive tracking.
- Request feedback from at least 5 hiring professionals on clarity versus novelty.

- Do not claim direct hiring improvement without observed evidence.

Commercial and operational risks

- Game navigation may slow busy recruiters or fail on corporate devices.
- Heavy media may hurt performance and accessibility.
- Confusing presentation could overshadow demonstrated backend skills.
- Time investment could displace applications and portfolio upkeep.

Phased go-to-market roadmap

- Launch standard portfolio with project PDFs first.
- Build game scene as an optional companion experience.
- Instrument task completion and optimize discoverability.
- Connect all RPG buildings to the same source-of-truth project data.

Sponsor / investor diligence checklist

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